SS-B-675b
February 8, 1968
SUPERSEDING
Fed. Spec. SS-B-675a
October 9, 1961

FEDERAL SPECIFICATION

BRICK, RUBBING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers abrasive rubbing bricks.

1.2 Classification.

1.2.1 Types and classes. Rubbing bricks shall be of the following types and classes, as specified (see 6.2):

Type I - Plain brick.
Type II - Fluted brick.
Class 1 - Without handle.
Class 2 - With handle.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Box, Fiberboard.
PPP-T-60 - Tape: Pressure Sensitive Adhesive, Waterproof, for Packaging.

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Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of Commerce:

CS 271 - Grading of Abrasive Grain for Grinding Wheels.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

American Trucking Associations, Inc.:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N. W., Washington, D. C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The rubbing bricks shall conform to figure 1, and as specified herein.

3.2 Material. Material shall be as specified herein. Material not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

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3.3 Grain and grade (see 6.3).

3.3.1 Grain. The grain shall be silicon-carbide abrasive, the numerical size (see 6.3.1) shall conform to CS 271, and shall be in accordance with table I as specified (see 6.2).

3.3.2 Grade (hardness). The grade of hardness of the brick shall be as specified in table I (see 6.2). The identification symbol of the degree of hardness is the bondage between the grains of abrasive, listed from G (soft) to U (hard) (see 6.3.2).

TABLE I. Identification symbols

Grit size		Grade		
		Soft	Medium	Hard
Fine	60, 70, 80	G, H, I	J, K, L, M	N, O, P, Q
Medium	30, 36, 46, 54	J, K, L	M, N, O, P	O, R, S, T
Coarse	16, 20, 24	L, M, N	O, P, Q	R, S, T, U

3.3.3 Bond. All rubbing bricks shall be vitrified bonded.

3.4 Surface. The surface shall be flat within 0.008 inch per lineal inch. Adjacent sides shall be square to one another within 5 degrees; opposite sides of the brick shall be parallel within 1/16-inch.

3.5 Identification marking. Each brick shall be identified in accordance with MIL-STD-130. In addition each brick shall be legibly marked with the grain size and grade symbol.

3.6 Dimensions. Dimensions of the bricks shall be in accordance with table II, as specified (see 6.2):

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TABLE II. Dimension

Type	Class	Thickness ± 3/16	Width ± 3/16	Length +3/8 -1/4
		Inches	Inches	Inches
I		1	2	6
		2	3	6
		2	2	8
		3	3	8
II	1	2	2	6
		2	2	8
		3	3	8
	2	1-1/2	3-1/2	8

3.7 Type I. The type I rubbing brick shall be a plain rubbing brick without handle or flutes.

3.8 Type II bricks. The type II rubbing brick shall have transversed flutes. The flutes shall extend at an angle from 12 degrees to 25 degrees to the longest edge of the brick. The flutes shall be not less than 3/8-inch wide and at least 1/4-inch deep. The width of the flutes may be tapered to not less than 1/4 inch at the deepest point. The center distance between flutes shall be not greater than 1-3/8 inch. The flutes shall not extend closer to the end of the brick than 3/16 inch. The end flutes may be blind, provided there is a wall of material at least 3/16-inch thick between the blind end of the flute and the end of the brick. All other flutes shall extend the entire width of the brick.

3.8.1 Class 1. The class 1 rubbing brick shall be without handle. Two opposing surfaces parallel to the longitudinal axis of the rubbing brick shall have flutes.

3.8.2 Class 2. The class 2 rubbing brick shall be provided with a handle. The widest surface shall be provided with flutes. The wooden handle shall be attached to the opposite surface to the flutes.

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3.8.2.1 Handle. The handle shall be made from a single piece of wood and shall be shaped so as to provide a comfortable handgrip. All corners shall be neatly rounded and shall have a smooth finish. The handle shall be 1/2 inch, plus or minus 1/4 inch, shorter than the long dimension of the brick and shall be attached to the brick by either cementing with waterproof cement or by wood screws. If wood screws are used, the screw heads shall be recessed not less than 1 inch below the rubbing surface of the brick, and a minimum of two wood screws shall be used.

3.9 Workmanship. The rubbing bricks shall be free from chips, voids or cracks. Distortion shall be limited to the tolerances specified. The handle, when so equipped, shall be free of nicks or splinters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Lot. All bricks of the same type and class offered for delivery at one time shall constitute a lot.

4.3.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.3 Examination. Samples selected in accordance with 4.3.2 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

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4.4 Inspection procedures.

4.4.1 Examination. The rubbing bricks shall be examined for the following defects:

101. Material not as specified.
102. Surface not as specified.
103. Handle not as specified.
104. Marking not as specified.
105. Dimension not as specified.
106. Type and class not as specified.
107. Workmanship not as specified.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108. Materials, methods and containers not as specified for level A or B. Each incorrect material, method or container shall constitute one defect.
109. Preservation not as specified for level A.
110. Packaging not as specified for level A.
111. Packing not as specified.
112. Strapping not zinc coated for level A.
113. Brick types and classes not segregated.
114. Marking illegible, incorrect or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A. The bricks of like types and classes shall be preserved in accordance with MIL-P-116, method III, using a box conforming to PPP-B-636, W5s or W5c, style optional, as the container. The box shall be closed in accordance with the box specification and secured with waterproof tape conforming to PPP-T-60, class 1.

5.1.2 Level C. Preservation and packaging of the brick shall be limited to that protection deemed necessary by the supplier to provide protection against deterioration and damage from the supplier to the initial destination for immediate use.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Bricks of like types and classes, preserved and packaged as specified in 5.1.1, shall be packed in close-fitting wood cleated-plywood boxes, conforming to PPP-B-601, overseas type, style optional. The boxes shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be zinc coated.

5.2.2 Level B. Bricks of like type and class, preserved and packaged as specified in 5.1.1, shall be packed as specified in 5.2.1, except the boxes shall be domestic type and strapping is not required to be zinc coated.

5.2.3 Level C. Bricks shall be packed in containers to insure carrier acceptance and safe arrival at destination. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required, interior and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use.

6.1.1 Type I. Type I plain rubbing bricks are used for foundry or machine shop work on castings or metals.

6.1.2 Type II. Type II fluted rubbing bricks are used by concrete or stone workers for rubbing down and finishing operations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Numerical grain size required (see 3.3.1).
(d) Alphabetic grade designation required (see 3.3.2).
(e) Dimensions required (see 3.6).
(f) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Grain and grades. These markings are for identification only, and not to indicate grinding action. Bricks bearing the same standard marking if made by different manufacturers may not and probably will not produce the same grinding action.

6.3.1 Grain size. The sizes of the abrasive grain shown in table I are listed to provide a size range enabling purchasing activities to select grit size based on the operation involved. However, type I plain rubbing bricks are generally furnished in grit numbers 20 coarse, 36 medium, and 80 fine and the type II fluted rubbing bricks are generally furnished in either grit number 16 or 20.

6.3.2 Grade. The abrasive products are manufactured in grades A through Z. The softest grade in which rubbing bricks are either commercially available or useful is G grade. On the other end of the range, grade U is normally the extreme of hardness utilized. Therefore the effective grade range is G through U. Furthermore the grit size greatly affect the grade normally used. As a general rule the finer the grit, the softer the stone.

MILITARY INTERESTS:

Custodians:
Army - ME
Navy - SH
Air Force - 84

Review activities:
Navy - SH
Air Force - 84

User activity:
Army - CE

Preparing activity:
Army - ME

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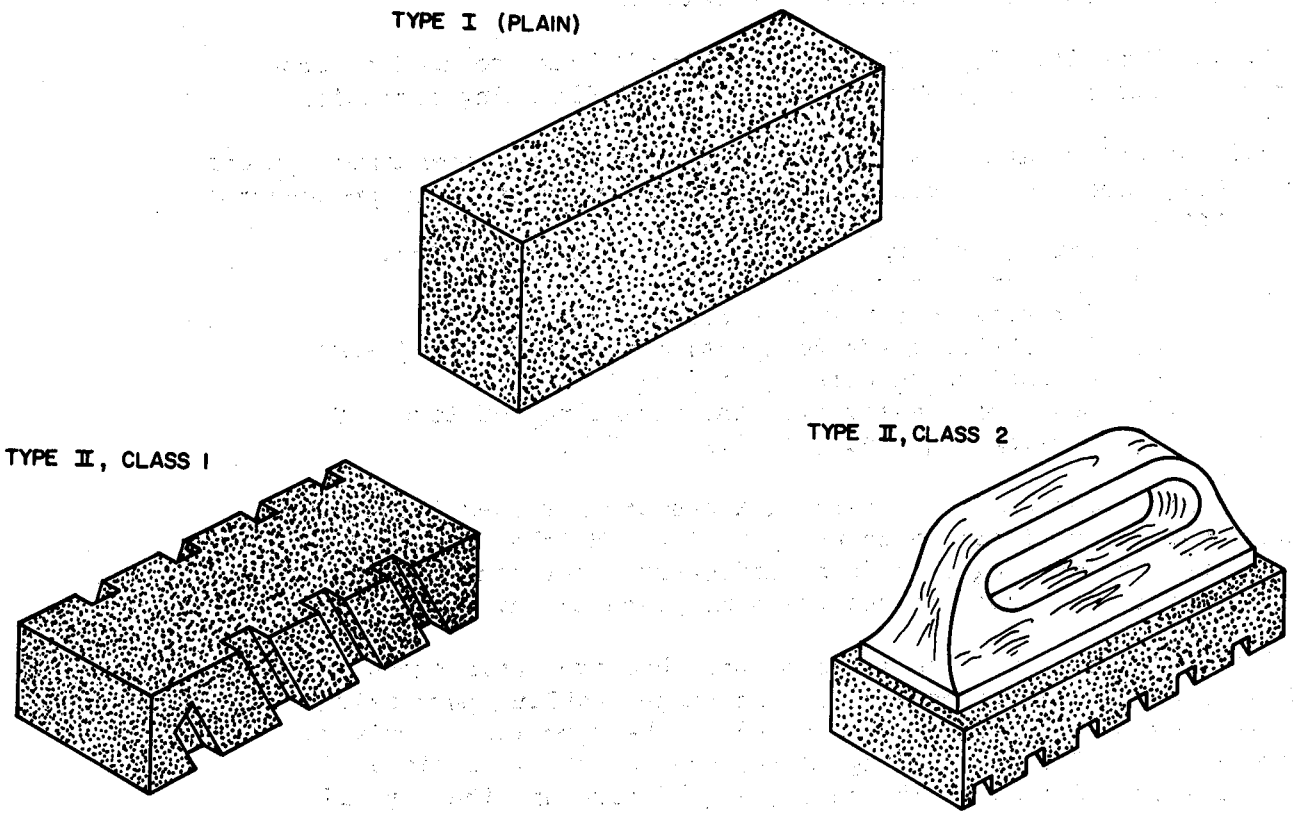


Figure 1. BRICK : RUBBING.

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